

FIND-Lab: A Factor-Controlled Benchmark for Informed Navigation Diagnosis in Constrained Spaces

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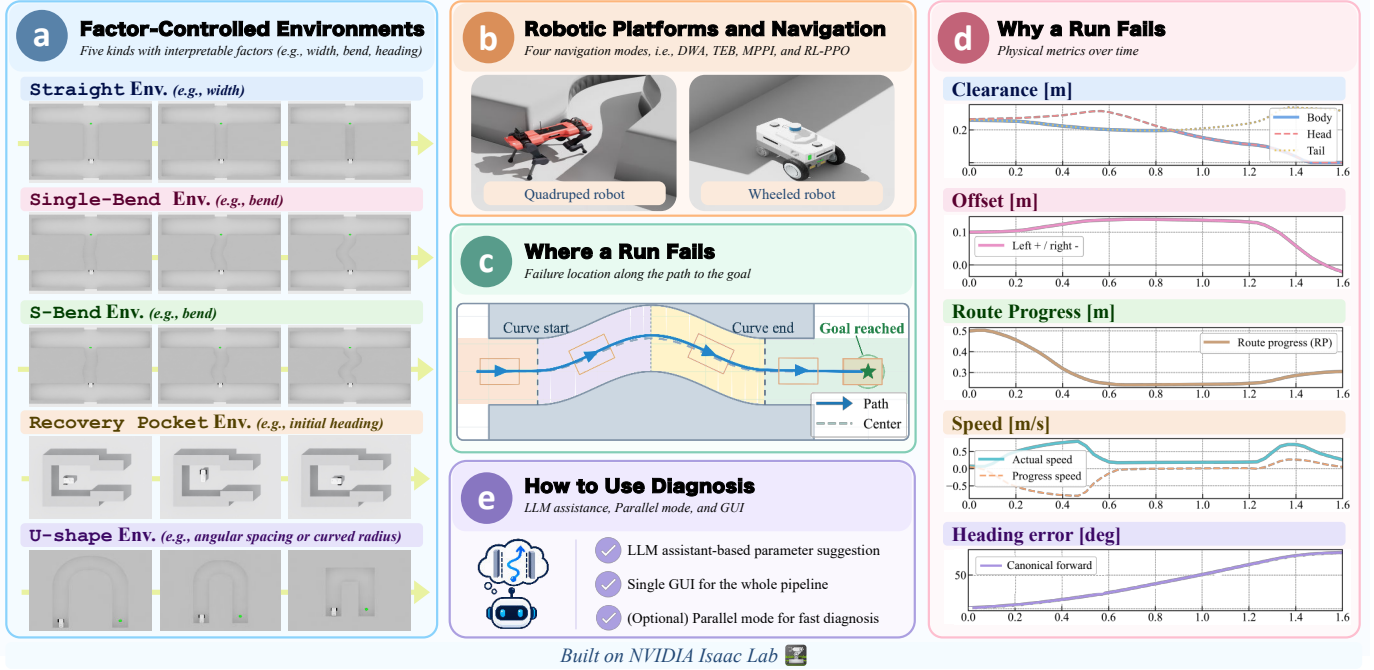


Fig. 1: Overview of *FIND-Lab*, built on Isaac Lab [1], linking controlled environments to run outcomes (*whether*), termination locations (*where*), diagnostic evidence (*why*), and parameter adjustment (*how*). (a) Five environment kinds, Straight, Single-Bend, S-Bend, Recovery Pocket, and U-Shape, vary passage width w , bend θ , initial heading ψ_0 , angular spacing d , and curved radius r . Settings vary from left to right; U-Shape has angular and curved variants. (b) Navigation methods include the dynamic window approach (DWA) [2], timed elastic band (TEB) [3, 4], model predictive path integral control (MPPI) [5], and reinforcement learning with proximal policy optimization (RL-PPO) [6], evaluated on Nova Carter [7] and ANYmal-C [8]. (c) The robot’s **path** and the **centerline** locate run termination, with **peach**, **purple**, **yellow**, and **green** marking entry, first bend, second bend, and exit. The schematic shows a run reaching the goal. (d) Time series show chassis clearance for the **body**, **head**, and **tail**, **centerline offset**, **route progress**, **actual** and **progress** speeds, and **heading error** relative to the route. (e) A large language model (LLM) reads diagnostic records and settings to propose parameter changes for re-evaluation. Furthermore, we provide a single graphical user interface (GUI) that covers (a)–(e) end to end, with optional parallel evaluation.

Abstract—Autonomous mobile robots are increasingly deployed in constrained spaces, such as narrow aisles and tight corners, where navigation often fails. Aggregate metrics in existing benchmarks mainly indicate *whether* a run fails, offering limited guidance on what to inspect or adjust. We present *FIND-Lab*, a factor-controlled benchmark for informed navigation diagnosis in constrained spaces. We first build five environment kinds in Isaac Sim, varying interpretable factors such as passage width and bend while holding other conditions fixed, so that every outcome is tied to the setting that produced it. Next, *FIND-Lab* shows *where* failed runs terminate and links these locations to behavioral time series of motion, clearance, and route alignment to investigate *why*. Finally, a large language model (LLM) uses these records and settings to suggest *how* to adjust planner parameters or policy rewards for re-evaluation. Notably, evaluations of four navigation methods across 33 factor settings on a wheeled robot (Nova Carter) and a legged robot (ANYmal-C) locate failure boundaries set by a configured margin rather than an impassable width, at a 0.30 m clearance penalty and at a 0.55 m obstacle envelope. In addition, among failures ending at similar locations, the records distinguish stalling with remaining clearance from continued motion until clearance is lost. Furthermore, across 29 Nova Carter conditions with reference success rates $< 50\%$, proposals informed by added diagnostics raise pooled success from 12.11% to 22.11% for reinforcement learning with proximal policy optimization over 19 conditions, from 1.25% to 18.75% for model predictive path integral control over 8, and from 30% to 40% for the dynamic window approach

over 2. We conclude that tying each outcome to its factor setting, termination location, and preceding behavior turns a failure count into evidence for what to inspect and adjust. Our project page is available at <https://findlab-1.github.io/FINDLab/>.

I. INTRODUCTION

Mobile robots perform repetitive, physically demanding logistics [9, 10], delivery [11], and inspection [12] tasks in factories, warehouses, restaurants, and hotels. These environments include narrow aisles, tight corners, and confined pockets where navigation failures occur [13, 14], making systematic evaluation and diagnosis essential for reducing failures.

Navigation benchmarks [15–17] address this evaluation need. The benchmark for autonomous robot navigation (BARN) dataset [17] orders simulated environments by difficulty (e.g., distance to the closest obstacle). Its annual challenge [18–22] has evaluated navigation systems using aggregate success rate and traversal time since 2022. These metrics summarize navigation success (*whether*) and traversal efficiency, enabling comparisons across systems.

However, diagnosing an individual failure requires information beyond these summaries, such as its termination

location (*where*) and preceding behavior as diagnostic evidence (*why*). Navigation couples perception, planning or a learned policy, and low-level control. Similar outcomes can mask different interactions with surrounding boundaries: a timeout can follow stationary behavior or repeated back-and-forth motion with little net progress [13, 18]. Aggregate metrics therefore offer limited guidance for inspecting or adjusting navigation parameters.

We present  *FIND-Lab*, a factor-controlled benchmark for diagnosing navigation failures in constrained spaces (Fig. 1). To compare behavior across controlled passage conditions, it links each run’s factor setting to its outcome (*whether*), termination location (*where*), and behavioral time series as diagnostic evidence (*why*). A large language model (LLM) then uses these records and navigation settings to propose parameter adjustments (*how*). Our contributions are:

- We present five constrained environment kinds (Straight, Single-Bend, S-Bend, Recovery Pocket, and U-Shape), controlling passage width, bend, angular spacing or curved radius, and the robot’s initial heading.
- We link termination locations along the centerline (*where*) to behavioral time series for interpreting stalls, rotations in place, retreats, oscillations of route progress, and clearance loss (*why*).
- In one ANYmal-C case and a pooled evaluation over 29 Nova Carter conditions, an LLM uses diagnostic records and settings to propose planner-parameter or policy-reward changes, which we re-evaluate (*how*).
- Across factor settings on wheeled and legged platforms [7, 8], four navigation baselines [2–6] place two failure boundaries at a configured margin rather than at an impassable width, one at a policy’s clearance penalty and one at a planner’s obstacle envelope.

Note that *FIND-Lab* is built on Isaac Lab [1] and evaluates its environments in parallel, which shortens a factor sweep and speeds up training, and that a single graphical user interface (GUI) covers the whole pipeline from building environments to visualizing their diagnosis. We hope our *FIND-Lab* helps users investigate differences in failure behavior and assess candidate parameter adjustments.

II. RELATED WORK

Table I compares benchmark environments and reported navigation performance and behavior. Based on this comparison, we review environment configuration in Sec. II-A, followed by evaluated methods, platforms, and outputs in Sec. II-B.

A. Construction of Constrained Environments

BARN [17] generates cluttered environments using cellular automata, and DynaBARN [15] varies object counts and motion profiles. Arena-Bench [16] supports adjustable corridor widths and configurable scenes. MRPB [23] uses manually designed scenes, including narrow passages with continuous U-shaped or acute-angle turns. These benchmarks set difficulty by what fills a space, by corridor width, or by scenes designed as a whole. Motivated by the passages of MRPB, *FIND-Lab* instead

Table I: Comparison of navigation benchmarks by their evaluation target, the geometric factors varied in their environment configurations, and their reported evaluation output. W and L denote wheeled and legged robots, respectively.

Benchmark	Evaluation target	Environment configuration	Reported evaluation output
BARN [17–22]	Local planner (W)	Object	Time, success
DynaBARN [15]	Local planner and policy (W)	Object	Success
MRPB [23]	Local planner (W)	N/A	Time, clearance, motion
Arena-Bench [16]	Local planner and policy (W)	Object, width	Time, success, clearance, motion
FIND-Lab (Proposed)	Local planner and policy (W, L)	Width, bend, initial heading, angular spacing, curved radius	Time, success, clearance, motion, progress, termination location

parameterizes the passage itself, sweeping width, bend, angular spacing, curved radius, and initial heading one factor at a time while other settings stay fixed.

B. Evaluated Local Planner, Platform, and Output

1) *Evaluated local planner*: These benchmarks evaluate local planners and learned navigation policies. BARN [17] compares the dynamic window approach (DWA) [2] against elastic bands (E-Band), and MRPB [23] compares DWA against timed elastic band (TEB) [3, 4], while DynaBARN [15] and Arena-Bench [16] add learned policies to the comparison.

2) *Robot platform*: These benchmarks use wheeled bases: a Clearpath Jackal in BARN [17] and DynaBARN [15], a differential-drive model with a circular footprint of radius 0.17 m in MRPB [23], and a Jackal, a TurtleBot3, or a Robotino in Arena-Bench [16].

3) *Reported evaluation output*: BARN [17] reports traversal time normalized by path length and assigns a fixed time penalty to a run that never reaches the goal, so failed and slow runs are summarized by the same metric. The BARN Challenge [18–22] reports success rate separately for its physical finals. DynaBARN [15] reports the success rate alone. MRPB [23] reports aggregate metrics for safety, efficiency, and smoothness based on the pose, velocity, and obstacle distance it logs at every planning call. A run that does not reach the goal leaves an empty entry, so how far it progressed is not reported either. Arena-Bench [16] adds collision count and obstacle clearance, both aggregated over each run and reported against object count rather than time.

Existing benchmarks offer diverse environments and aggregate time, success, clearance, or motion metrics for wheeled robots. *FIND-Lab* links controlled passage factors to outcomes, termination locations (*where*), and behavioral time series for investigating *why*. It compares wheeled and legged systems under the same factor settings, providing evidence that may inform system inspection and parameter adjustment (*how*).

III. FROM OBJECTS TO PASSAGE GEOMETRY

A constrained space offers little clearance relative to the robot’s dimensions. Even a small deviation from the intended motion can therefore bring the robot into contact with a boundary [13, 14, 18]. At some tested factor settings, most runs end in collision, timeout, or other failures. We first narrow

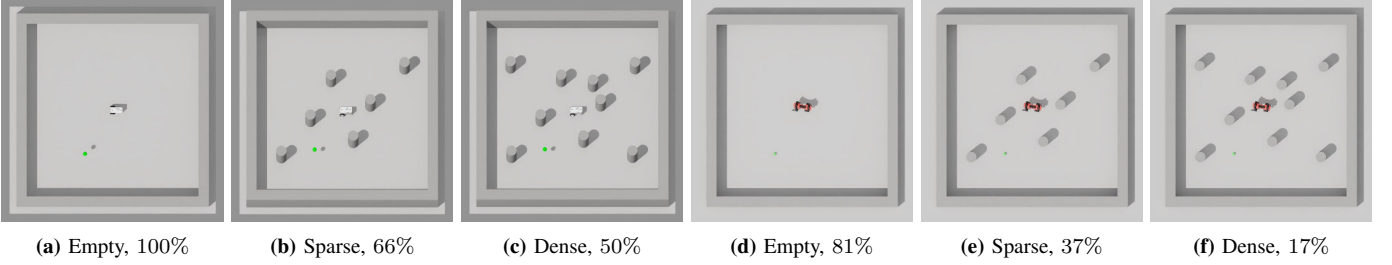


Fig. 2: Adding objects reduces navigation success on both platforms in Isaac Sim with Isaac Lab [1]. The same reinforcement learning policy trained with proximal policy optimization (RL-PPO) [6] on Nova Carter [7] is evaluated unchanged on Nova Carter in (a)–(c) and, with its commands transformed for the existing locomotion controller, on ANYmal-C [8] in (d)–(f). Each percentage reports successes out of 100 runs: reaching within 0.3 m of the goal in 30 s without collision or fall. The ANYmal-C failures include falls, discussed in Sec. III-A.

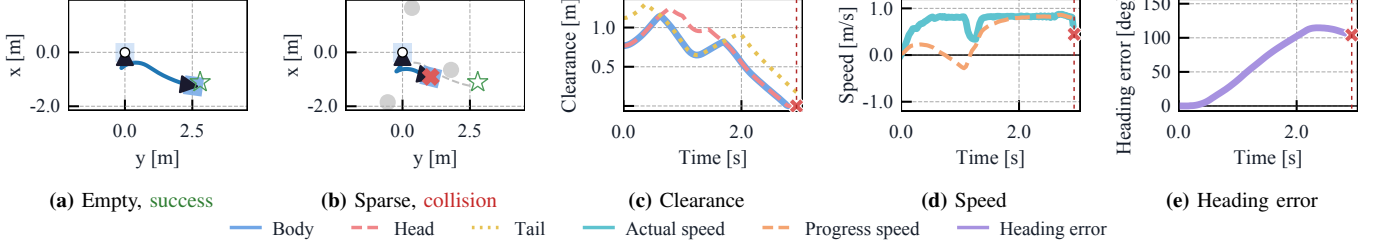


Fig. 3: The same reinforcement learning with proximal policy optimization (RL-PPO) [6] policy, start, and goal on Nova Carter [7] produce success in the empty room (a) and collision in the sparse room (b). Success occurs at 4.67 s, and $\times$ marks the collision at 2.93 s, 1.75 m from the goal, with red dotted lines at the same time in (c)–(e). Gray disks are objects, the dashed gray path repeats (a), and stars mark the goal. Panels (c)–(e) show the failed run: chassis clearance, **actual speed** and **progress speed**, and absolute **heading error** between the robot’s physical forward direction and the reference route.

a space with scattered objects in Sec. III-A, then examine passage geometry in Sec. III-B.

A. Narrowing a Space with Objects

The benchmarks of Sec. II narrow a space by what they place in it, so we begin the same way. We use a fixed reinforcement learning checkpoint trained with proximal policy optimization (RL-PPO) [6] to illustrate basic navigation capability and its sensitivity to added objects. Fig. 2 evaluates this checkpoint in one room under empty, sparse, and dense object layouts, with 100 runs per layout on each platform. The training conditions and hyperparameters are described in Sec. IV. As the objects are added, the free space between them narrows and the success rate falls from 100% to 66% and 50% on Nova Carter [7] in Fig. 2(a)–(c), and from 81% to 37% and 17% on ANYmal-C [8] in Fig. 2(d)–(f). The ANYmal-C rates include falls rather than navigation failures alone: all 19 empty-room failures were falls, and a separate evaluation with a velocity limiter succeeded in all 100 episodes. These rates show *whether* runs succeed, but omit termination location, motion, and surrounding clearance.

Fig. 3 compares the same policy succeeding in an empty room and colliding among scattered objects. The trajectories in Fig. 3(a)–(b) locate termination: success at 4.67 s versus collision at 2.93 s, 1.75 m from the goal. The time series in Fig. 3(c)–(e) provide evidence to investigate *why*. In Fig. 3(d), the robot maintains about 0.80 m/s for almost the whole run, yet its progress speed becomes negative between 0.77 and 1.17 s, indicating a temporary decrease in route progress. It then maintains its actual speed until contact as the head clearance of Fig. 3(c) falls below 0.3 m from 2.33 s onward. The heading error in Fig. 3(e) describes alignment of the robot’s physical forward direction with the reference route. At contact, the head and body clearances reach zero while the tail clearance remains

0.18 m. This evidence may inform *how* a parameter is adjusted, which we examine with an LLM-based assistant in Sec. V.

B. Beyond Objects

Equal object densities can yield different passage widths, bends, and turn spacing. We therefore parameterize passage geometry directly for systematic variation, covering bent aisles, successive bends, tight corner pairs, and confined pockets requiring reorientation. These describe the passage itself rather than the objects within it.

Passage width w is the distance between the inner boundaries of the two walls, interpreted relative to robot width b , the lateral width of its planar footprint. Here, $b = 0.70$ m for Nova Carter and $b = 0.55$ m for ANYmal-C. The bend θ sets how the passage turns. Successive right-angle corners are separated by angular spacing d , a distance, while curved turns use radius r ; ψ_0 sets the initial heading offset. Table II lists these factors by environment kind. Our *FIND-Lab* systematically varies passage geometry and initial heading across five environment kinds, linking each factor setting to run outcomes, termination locations, and behavioral time series.

IV. BENCHMARKING NAVIGATION ON FIND-LAB: WHETHER, WHERE, AND WHY

A. Benchmark Setup

All experiments used Isaac Sim 5.1.0 [24] with Isaac Lab 2.3.2 [1] with four NVIDIA RTX A6000 GPUs.

1) *Robot platforms and sensing*: We evaluate the wheeled Nova Carter [7] and the legged ANYmal-C [8]. Each robot uses idealized 360° raycasting with its NVIDIA Isaac Sim 5.1 asset, with ranges clipped to 0.45–6 m for the mesh raycast and 0–100 m for the VLP-16 raycast. All methods on Nova Carter and RL-PPO on ANYmal-C use 36-bin range observations,

Table II: Environment kinds, swept factors, and diagnostic focus. w : passage width; θ : bend (the lateral shape of the route); ψ_0 : initial heading (the offset from the exit direction); d : angular spacing, measured as the distance between two right-angle corners; and r : curved radius of the turns. U-Shape has angular and curved variants. Each sweep varies the listed factor alone; Single-Bend and S-Bend are also parameterized by w , which is held fixed at 1.2 m while θ is swept.

Environment	Swept factor	Diagnostic focus
Straight	width w	Entry and sustained clearance
Single-Bend	bend θ (fixed w)	Alignment through a curved section
S-Bend	bend θ (fixed w)	Alternating turns and realignment
Recovery Pocket	initial heading ψ_0	Reorientation and escape
U-Shape	angular spacing d or curved radius r	Progress and clearance through successive turns

and classical planners on ANYmal-C use a 16-channel VLP-16 raycast. The two configurations differ because the pretrained PPO observation interface is preserved, while classical planners on ANYmal-C retain their existing Isaac Lab VLP-16 sensing configuration.

2) *Environments and controlled factors*: We vary passage geometry across five environment kinds, whose swept factors and diagnostic focus are summarized in Table II. Scene geometry provides the obstacle polygons and the centerline used for diagnosis.

3) *Navigation baselines*: We evaluate three classical planners, DWA [2], TEB [3, 4], and model predictive path integral control (MPPI) [5], and a learned RL-PPO policy [6] on both robots (settings in Table III). All methods use shared route guidance derived from the centerline, without a global planner. The classical planners use our DWA implementations, the upstream `teb_local_planner` single-band optimizer (v0.9.1), and the Nav2 MPPI controller (ROS 2 Jazzy), respectively.

RL-PPO was trained on Nova Carter in 16 parallel copies of a single scene of scattered objects, with goals 0.8–3.0 m away. Its reward includes a wall-clearance penalty that is active only below a chassis-to-wall clearance of 0.30 m. The evaluated checkpoint was selected using the success rate over the last 100 training episodes without a held-out split. On ANYmal-C, commands from the same checkpoint are transformed for the existing locomotion controller. Neither the benchmark environment kinds nor ANYmal-C was used in training, so the evaluation tests transfer across geometries and platforms.

4) *Evaluation protocol*: We sweep one factor while fixing all other geometry, start and goal placement, sensing, route guidance, and navigation settings. Six sweeps (separating angular and curved U-Shape) comprise 33 factor bins, each evaluated from ten fixed start poses shared by both robots and all four methods: $33 \times 10 \times 2 \times 4 = 2,640$ episodes. Planner-internal sampling is unseeded, so stochastic methods such as MPPI are not bitwise reproducible, whereas the episode initialization seeds are fixed and shared.

5) *Run outcomes and failure cases*: Success requires reaching within 0.30 m of the goal in the plane within 30 s of simulated time, without collision or, for ANYmal-C, a fall. Other runtime outcomes are collision, timeout, planner failure,

Table III: Settings for the dynamic window approach (DWA) [2], timed elastic band (TEB) [3, 4], model predictive path integral control (MPPI) [5], and reinforcement learning with proximal policy optimization (RL-PPO) [6]. All methods share maximum commanded linear and angular speeds and the navigation command rate. The obstacle envelope is the minimum distance required between a candidate robot reference point and obstacle points.

Method	Parameter	Value
Common	Max. linear / angular speed	0.8 m/s / 1.2 rad/s
	Navigation command rate	30 Hz
DWA [2]	Simulation horizon, velocity samples	2.0 s, 6 / 20
	Goal / path / obstacle cost weights	1.0 / 0.75 / 0.10
	Obstacle envelope on ANYmal-C	0.55 m
	Obstacle envelope on Nova Carter	0.43 m
TEB [3, 4]	Min. obstacle distance	0.10 m
	Reference time step	0.30 s
	Optimization iterations	5 inner / 4 outer
MPPI [5]	Samples, prediction horizon	2,000, 84 steps (2.8 s)
	Prediction time step, temperature	1/30 s, 0.3
	Noise std. (linear / angular)	0.2 m/s / 0.4 rad/s
RL-PPO [6]	Fixed checkpoint	3.58×10^6 transitions

Table IV: Behavioral patterns in the logged time series. Run outcomes are recorded separately: **timeout**, 30 s elapsed; **collision**, horizontal contact-sensor force > 100 N; **success**, planar goal distance ≤ 0.30 m within 30 s without collision or fall. A **fall** is an ANYmal-C base height below 0.32 m above the ground.

Behavioral pattern	Metric evidence
Stall	$v_k \approx 0$ and $v_{s,k} \approx 0$, with negligible angular motion in unwrapped ψ_k
Rotation in place	$v_k \approx 0$ and $v_{s,k} \approx 0$, with substantial angular motion in unwrapped ψ_k
Retreat along route	Decreasing s_k with $v_{s,k} < 0$ over a sustained interval
Oscillation of route progress	Repeated substantial reversals in s_k , with small absolute net change relative to the total absolute change in s_k
Clearance loss	d_k^{body} , d_k^{head} , or d_k^{tail} decreasing to near zero, with $e_{\perp,k}$ and $e_{\psi,k}$ as alignment context

and ANYmal-C fall. The main evaluation runs with the velocity limiter of Fig. 2, so a recorded fall is not one the limiter removes. Collision is detected when the horizontal force measured by a body contact sensor exceeds 100 N. Per-cycle candidate invalidity is recorded separately from terminal planner failure. Reset rejections, in which the robot cannot maintain its initial pose before navigation, are counted separately.

Table IV lists qualitative behavioral patterns used to interpret failed runs. The metrics used as evidence are defined as follows:

- $s_k = \operatorname{argmin}_s \|\mathbf{p}_k - \gamma(s)\|_2$,
- $e_{\perp,k} = \mathbf{n}(s_k)^\top (\mathbf{p}_k - \gamma(s_k))$,
- $d_k^a = \min_{O \in \mathcal{O}} \operatorname{dist}(\mathcal{B}_k^a, O)$,
- $v_k = \|\mathbf{v}_k^{\text{world}}\|_2$,
- $v_{s,k} = (s_k - s_{k-1}) / (t_k - t_{k-1})$,
- $e_{\psi,k} = |\operatorname{wrap}_{[-180^\circ, 180^\circ]}(\psi_k - \alpha(s_k))|$,

where s_k [m] is the route progress and $e_{\perp,k}$ [m] the lateral offset of an independent per-step projection onto the centerline, positive to the left; d_k^a [m] is the clearance of the chassis envelope for its $a \in \{\text{body}, \text{head}, \text{tail}\}$ regions; v_k [m/s] is the actual speed; $v_{s,k}$ [m/s] is the progress speed; and $e_{\psi,k}$ [deg] is the absolute heading error between the physical forward direction and the local route direction. Here, $\mathbf{p}_k \in \mathbb{R}^2$, ψ_k , and $\mathbf{v}_k^{\text{world}}$ are the logged position, physical forward heading, and measured planar velocity at time t_k ; $\gamma(s)$ is the piecewise-linear

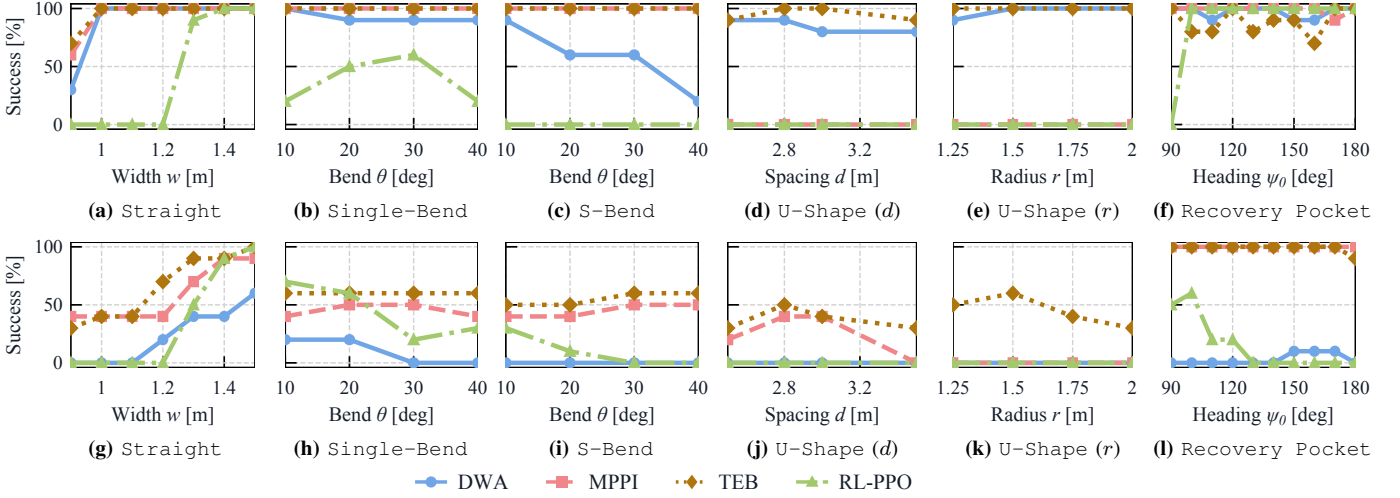


Fig. 4: Success rates depend on both passage geometry and robot platform, with Nova Carter [7] above in (a)–(f) and ANYmal-C [8] below in (g)–(l). Columns sweep passage width w , bend θ for Single-Bend and S-Bend, angular spacing d , curved radius r , and initial heading ψ_0 . Each marker is the descriptive success rate over the same $n = 10$ fixed start poses per factor setting; other factors remain fixed. Methods are the dynamic window approach (DWA) [2], timed elastic band (TEB) [3, 4], model predictive path integral control (MPPI) [5], and reinforcement learning with proximal policy optimization (RL-PPO) [6].

centerline parameterized by arc length with left-facing unit normal $\mathbf{n}(s)$ and route direction $\alpha(s)$; $\mathcal{B}_k^a$ is the rectangular chassis envelope with configured pose offsets; and dist is the Euclidean polygon distance to an obstacle polygon in $\mathcal{O}$, zero at contact or overlap. Note that the chassis envelope excludes ANYmal-C’s articulated legs, and zero chassis clearance does not replace the contact-sensor collision outcome. These patterns do not determine recorded outcomes, root causes, or which parameters to adjust. Interpretation depends on motion duration and magnitude; minor fluctuations or projection discontinuities do not qualify, and rotation or retreat may reflect recovery [25]. A termination location is the last valid pose, mapped to phases before, within, and after each passage or turn, and it need not mark where the problem began.

B. Whether: Outcomes across Factor Settings

Fig. 4 separates outcomes by environment kind and factor setting. Notably, two of its transitions fall where a setting, not the passage, stops admitting motion. Nova Carter RL-PPO fails in every Straight episode up to $w = 1.2$ m and succeeds in 90% at $w = 1.3$ m, and with $b = 0.70$ m these widths leave 0.25 and 0.30 m of wall clearance, so the transition coincides with the 0.30 m threshold of the policy’s own training penalty. ANYmal-C DWA likewise fails up to $w = 1.1$ m, below which the 0.55 m obstacle envelope of Table III leaves no admissible candidate, so both transitions mark a configured margin rather than an impassable width. In addition, factor settings separate methods within a platform. Nova Carter DWA succeeds in 90% of S-Bend episodes at 10° but 20% at 40° . Nova Carter MPPI succeeds in all tested S-Bend episodes but fails in both U-Shape variants. Recorded failures include 104 MPPI path-computation errors and 79 TEB errors from non-positive time steps, totaling 183 planner failures. All 49 reset rejections occur on ANYmal-C. Note that planner failures count as failed episodes, and reset rejections are resampled rather than dropped, so every factor setting retains ten evaluated episodes.

C. Where and Why: Termination Locations and Diagnostic Evidence

Notably, runs that end in the same passage region need not have behaved the same way beforehand (Fig. 5).

Five of the eight runs fail, and they terminate in three passage regions: ANYmal-C DWA near the start ($s_k = 0.01$ m), RL-PPO before the turn (1.07–1.40 m), and MPPI farther along (7.71–8.05 m). The two MPPI runs end near each other but differ in motion and clearance. On Nova Carter, s_k increases only 0.05 m over the final 10 s while $d_k^{\text{body}} \approx 0.32$ m, supporting an interpretation of stalled motion. Although $v_{s,k}$ changes sign there, the run retreats only 0.004 m per meter of forward progress, providing little evidence of substantial back-and-forth motion. ANYmal-C continues moving until clearance is lost at 21.7 s.

V. HOW: REVIEWING AND EVALUATING PARAMETER ADJUSTMENTS

Inspired by LLM-based reward design [26], failure explanation [27], adaptive planner parameter learning [28], and language-model-based planner tuning [29], we review parameter adjustments proposed with and without FIND-Lab diagnostic inputs and report their re-evaluation outcomes under the selected conditions. We compare aggregate (A) and added diagnostic (L) inputs in one S-Bend case and in a pooled evaluation on Nova Carter (Table V).

The case used ANYmal-C [8] with DWA [2] at $\theta = 10^\circ$. A and L received identical success, collision, and timeout aggregates from 40 original episodes per target, setting definitions, original values, and allowed changes. L additionally received prepared plots and factual summaries of a failure case fixed before the calls: termination location, progress, speed, clearance, offset, and heading error, plus DWA execution summaries. Each target used one gpt-6-astra call per condition with ultra reasoning effort and tools disabled, without access to other responses or re-evaluation results. Scripts checked that the proposed changes were permitted and applied correctly.

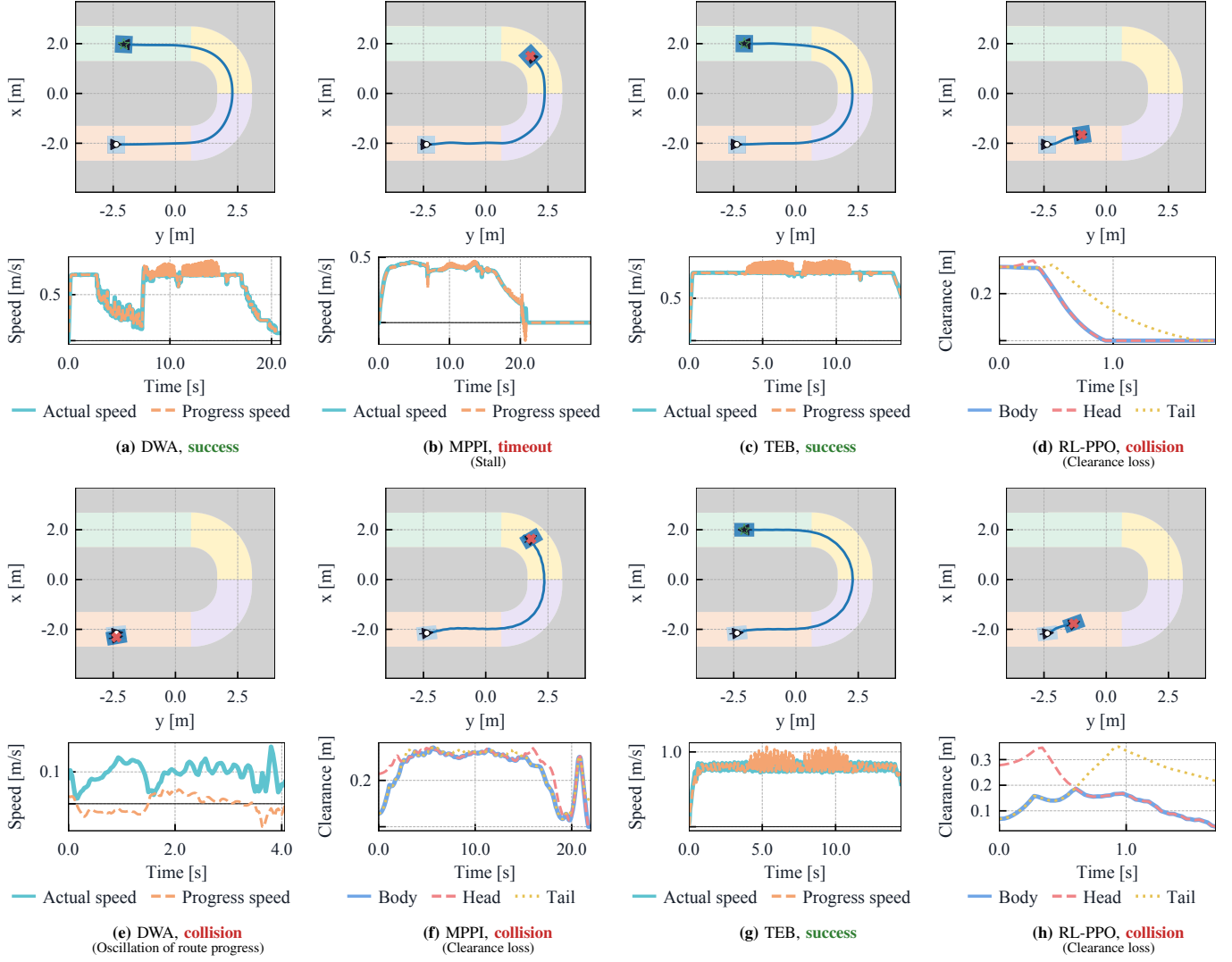


Fig. 5: Similar termination locations can hide different behavior in the curved U-Shape ($r = 1.75$ m), shown for one episode index shared by all eight runs. Nova Carter [7] (top, (a)–(d)) and ANYmal-C [8] (bottom, (e)–(h)) use the dynamic window approach (DWA) [2], model predictive path integral control (MPPI) [5], timed elastic band (TEB) [3, 4], and reinforcement learning with proximal policy optimization (RL-PPO) [6]. Maps show paths, light start and solid end footprints, heading arrows, stars for success, and crosses for timeout or collision. **Peach**, **purple**, **yellow**, and **green** denote entry, the two turn halves, and exit. Speed traces show measured planar speed (**actual speed**) and the rate of route progress (**progress speed**). Clearance traces show the minimum obstacle distance for the whole chassis envelope (**body**) and its front (**head**) and rear (**tail**) halves. Failure annotations use the behavioral patterns in Table IV.

Fig. 6 shows the two responses. With A, the LLM hypothesized insufficient planning margin and proposed increasing the obstacle envelope from 0.55 to 0.65 m. With L, the LLM interpreted the plotted combination of stalled progress near $s_k = 3.5$ m, growing centerline offset and heading error, and tail clearance approaching zero as suggesting difficulty recovering toward the route. It also used the separately provided DWA execution summary, which reported that 439/449 commands after 10 s lacked a valid trajectory, to support the hypothesis that the envelope restricted corrective motion. On this combined basis, it proposed reducing the envelope to 0.45 m, leaving physical collision detection unchanged.

F denotes the original settings or policy. For DWA, each proposal changed one permitted setting while holding the others fixed. F/A/L used matching goals, environment seeds, and initial positions, which vary within ± 0.1 m across the ten episodes of a condition while the goal stays fixed. For the case, these ten episodes are distinct from the 40 input episodes. MPPI and

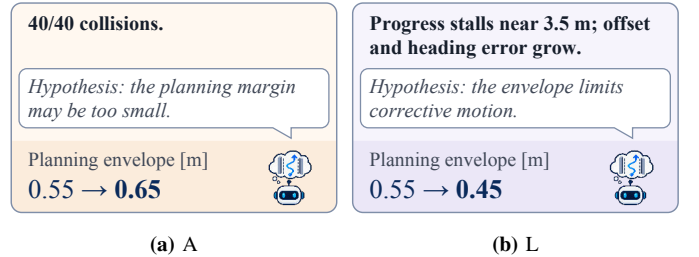


Fig. 6: Paraphrased observations, tentative interpretations, and applied settings from separate LLM calls for the case of Sec. V, ANYmal-C [8] with the dynamic window approach (DWA) [2]. (a) A uses outcome aggregates, setting definitions, original values, and allowed changes, and (b) L adds diagnostic plots and DWA execution summaries. Each card shows the hypothesis and the original-to-applied value, and neither establishes a failure cause.

DWA proposals were applied without training, new LLM calls, or retuning. Because each condition receives one LLM call per input, episodes within a condition are not independent tests of the input, and we therefore report pooled rates descriptively and do not test the A–L difference.

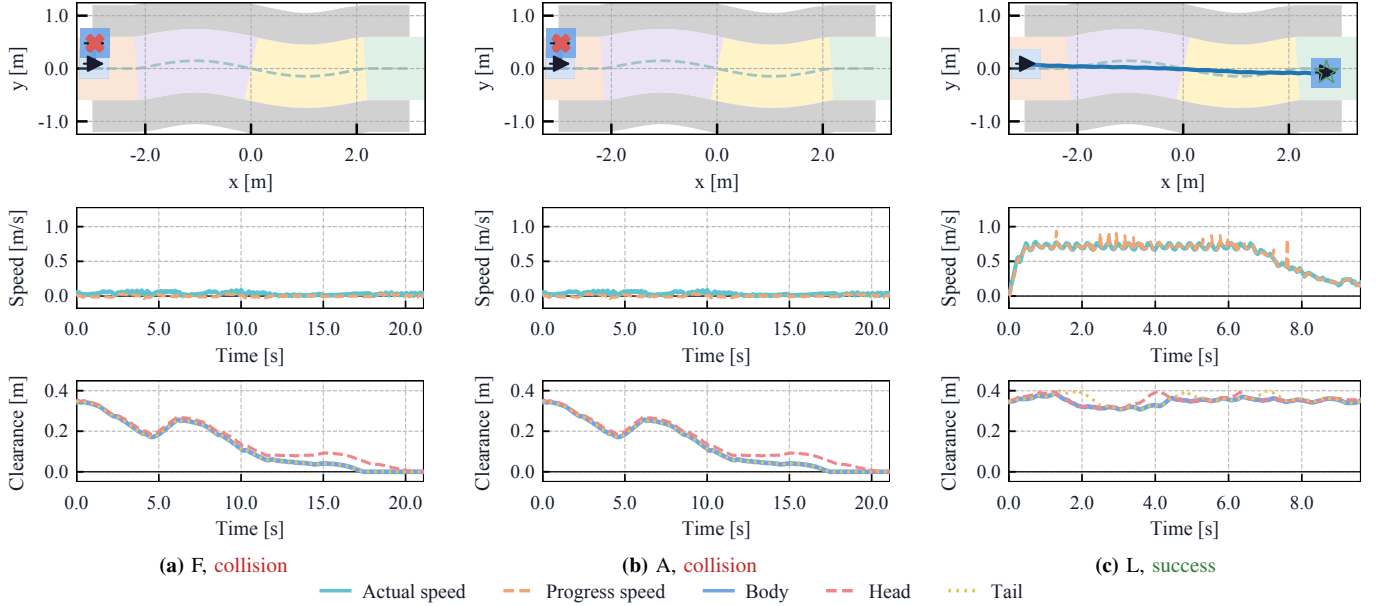


Fig. 7: One episode each from the case of Sec. V, ANYmal-C [8] with the dynamic window approach (DWA) [2]. (a) A recorded failure at the original settings, and (b), (c) re-evaluations with the settings proposed from the aggregate (A) and added diagnostic (L) inputs of Fig. 6. These are single episodes rather than a controlled before–after comparison, and the recorded failure in (a) is not the episode summarized for L. Maps show paths, dashed centerlines, light start and solid end footprints, and stars for success and crosses for collision. Peach, purple, yellow, and green mark entry, first bend, second bend, and exit. The middle row shows **actual speed** v_k and **progress speed** $v_{s,k}$, and the bottom row shows **body**, **head**, and **tail** clearance.

Table V: Pooled successes/episodes (success rate) on Nova Carter [7] conditions with reference success rates $< 50\%$ in Fig. 4. MPPI and DWA use ten episodes per condition, and RL-PPO reports 190 episodes per input. F: reference settings or policy; A: aggregate input; L: added diagnostics. Bold marks the highest success rate per method. Colours mark the 1st, 2nd, and 3rd result. F repeats the Fig. 4 values used to select these conditions, so gains over F include regression to the mean. RL-PPO A and L both include additional training, so the A–L difference isolates the input whereas gains over F do not.

Method	F (reference)	A	L
MPPI [5]	0/80 (0%)	1/80 (1.25%)	15/80 (18.75%)
DWA [2]	5/20 (25%)	6/20 (30%)	8/20 (40%)
RL-PPO [6]	4/190 (2.11%)	23/190 (12.11%)	42/190 (22.11%)

For each RL-PPO condition and input, training continued from the original policy for up to 50 iterations, with validation on four separate episodes every 10 iterations. After at least 30 iterations, training stopped once two consecutive validations showed no improvement in successes, collisions, or route progress, and the selected checkpoint was evaluated on ten episodes. Identical A/L proposals shared training and evaluation.

Fig. 7 shows one episode per setting, in which the robot collides early under F and A and reaches the goal under L. The case is consistent with testing a smaller envelope when the diagnostics suggest restricted corrective motion, without establishing the failure cause.

For Table V, we selected Nova Carter conditions with reference success rates $< 50\%$ in Fig. 4, applied A/L proposals, and pooled re-evaluation outcomes by method. We restricted the pooled evaluation to Nova Carter so that comparisons of navigation-level adjustments are not confounded by ANYmal-C’s learned locomotion controller, which lies outside the permitted changes. The case study is a single illustrative record rather than a controlled comparison, so it is not subject to

this restriction. The selection yielded 8 MPPI, 19 RL-PPO, and 2 DWA conditions, with ten episodes per condition and input. No Nova Carter TEB condition met this criterion, so TEB does not enter the pooled evaluation. These episodes reuse the Fig. 4 start poses, goals, and environment seeds, making the pooled evaluation an in-sample re-evaluation. For MPPI, each proposal could scale at most one critic weight of the Nav2 MPPI controller [25] by a factor of 0.05–20, holding all other settings fixed. For RL-PPO, both A and L adjusted the coefficient λ of the predefined wall-clearance penalty $-\lambda[\text{clip}((0.30 - c)/0.30, 0, 1)]^2$, where c is the chassis-to-wall clearance in meters. For MPPI and RL-PPO, L additionally received only FIND-Lab plots, numerical summaries, and metric definitions, without planner-internal error or candidate-validity summaries. From A to L, success rose from 1.25% to 18.75% for MPPI, 30% to 40% for DWA, and 12.11% to 22.11% for RL-PPO, gains of 17.5, 10, and 10 percentage points. These compare inputs within each method, since evaluation sets differ, and pooled gains do not imply improvement in every condition.

Together, the results support using diagnostics to propose and test adjustments. The added diagnostics most often changed the magnitude of an adjustment rather than its target, and they are associated with higher pooled success here, though they need not improve every case. In the ANYmal-C case they reversed the direction of the adjustment. The proposals establish neither failure causes nor optimal settings.

VI. DISCUSSION

A. Scope of Diagnostic Evidence

Success rates across factor settings (Fig. 4) help select conditions for closer inspection. Termination locations show where runs end, while time series (Fig. 5) reveal differences in

how the robots move before termination. RL-PPO and ANYmal-C DWA collide at different progress levels; MPPI stalls with clearance on Nova Carter but moves until clearance is lost on ANYmal-C. These observations guide investigation of the coupled perception–planning–control loop [13, 14]. A stall may reflect inadmissible plans, missed traversable space, or unrealized motion commands; a single record cannot identify the responsible component. Candidate invalidity per planning cycle also differs from the runtime episode outcome.

B. Scope of the Case Studies

FIND-Lab supports proposing and testing settings from supplied evidence. Across the 29 pooled conditions, A and L proposed identical changes in one, changed the same setting in the same direction by different amounts in 22, and differed in the selected setting or in whether to intervene in six. The most common difference was therefore the magnitude of the proposed adjustment. The numerical choices are exploratory trial values; neither the model’s hypotheses nor the values establish a failure cause or an optimum.

Because the diagnostic inputs were provided together, this evaluation does not isolate the contribution of individual metrics to these proposal differences. In the DWA case study, this also applies to the time series and execution summary, both of which were provided only to L. RL-PPO differences from F combine reward modification with additional training, so they do not isolate the reward contribution. With one call per experiment and condition and one RL-PPO training seed, conclusions are limited to these proposals and evaluation conditions.

VII. CONCLUSION

We presented  *FIND-Lab*, which varies interpretable factors across five constrained environment kinds and links outcomes (*whether*), termination locations (*where*), and behavioral time series as diagnostic evidence (*why*). Wheeled and legged evaluations [7, 8] identified challenging settings and distinct behaviors behind shared outcomes or termination regions. In one ANYmal-C case and a pooled evaluation over 29 Nova Carter conditions, diagnostics informed DWA and MPPI parameter proposals and RL-PPO reward proposals (*how*), which we re-evaluated. These records guide investigation and parameter adjustment, including cost weights, without identifying a responsible component. Future work will test physical robots under passage geometries and factor settings matched to simulation, assessing diagnostic support for failure investigation and improvement. It will also supply diagnostic records one at a time, and instrument the perception, planning, and control layers separately, so that a stall can be attributed rather than only described. Versioned releases will add environment kinds, factors, and baselines, keeping earlier settings reproducible to support navigation development across environmental constraints.

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